$$P(A \cup \bar{A}) = P(A) + P(\bar{A}) \text{ (see figure 8.4)}$$

$$P(S) = P(A) + P(\bar{A})$$

$$1 = P(A) + P(\bar{A}) \quad \text{(by Axiom-2)}$$

It implies that $P(\bar{A}) = 1 - P(A)$.

NOTE

Since $\bar{A}$ is an event, by Axiom -1, $P(\bar{A}) \geq 0$.
i.e., $1 - P(A) \geq 0$ $\quad P(A) \leq 1$.

Thus, the probability for the occurrence of any event is always a real number between 0 and 1 i.e., $0 \leq P(A) \leq 1$.

Theorem 8.3: If A and B are two events in an experiment such that $A \subset B$, then $P(B-A) = P(B) - P(A)$.

Proof:

It is given that $A \subset B$.

The event B can be expressed as

$B = A \cup (B-A)$ (see Figure 8.6)

Since $A \cap (B-A) = \varphi$,

$$P(B) = P(A \cup (B-A))$$

Hence, by Axiom-3,

$$\Rightarrow \quad P(B) = P(A) + P(B-A)$$

Therefore, $P(B-A) = P(B) - P(A)$.

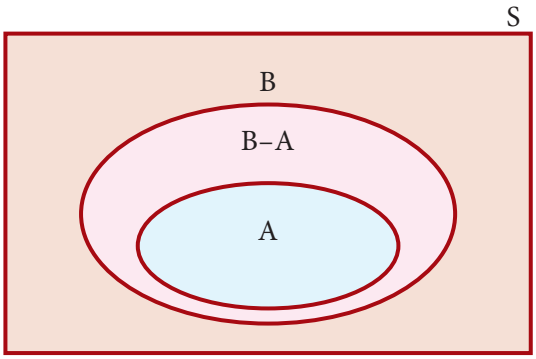


fig. 8.6 Venn diagram

Corollary: If $A \subset B$, then $P(A) \leq P(B)$.

Proof:

Since, by Axiom-1, $P(B-A) \geq 0$, it follows that

$P(B) - P(A) \geq 0$

$P(B) \geq P(A)$

$$\Rightarrow \quad P(A) \leq P(B).$$